

Experiment 1: LaTeX Implementation Source Code

Below is the exact and unaltered LaTeX code structure with original indentation, packages, and commands intact:

```
\documentclass{article}
\usepackage{fancyhdr}
\usepackage{lipsum}

\pagestyle{fancy}
\fancyhf{}

\fancyhead[R]{Latex Programming}
\fancyhead[L]{Vinyas.K.S, 4VV24CS047}
\fancyfoot[R]{\thepage}
\fancyfoot[L]{VVIET, Mysore}

\begin{document}

    \section{Section 1}
    \lipsum[1-2]

    \section{Section 2}
    \lipsum[1]

\end{document}
```

Experiment 2: LaTeX Implementation Source Code

Below is the exact and unaltered LaTeX code structure with original indentation, packages, and commands intact:

```
\documentclass{article}
\usepackage{fancyhdr}
\usepackage{lipsum}

\pagestyle{fancy}
\fancyhf{}
\fancyhead[R]{Technical Writing using LaTeX}
\fancyfoot[C]{\thepage}
\fancyfoot[L]{VVIET, Mysore}

\begin{document}

    \title{Technical Writing using LaTeX}
    \author{Vinyas K.S}
    \date{\today}
    \maketitle

    \thispagestyle{fancy}

    \begin{abstract}
        Technical writing is used to communicate technical information clearly
and effectively.
        LaTeX is a powerful document preparation system commonly used for
creating scientific papers,
        reports, and academic documents.
        It provides high-quality formatting and helps organize
documents in a professional manner.
    \end{abstract}

    \section{Introduction}
    Technical writing is important in engineering and research fields.
    It helps present ideas,
    experiments, and results in a structured format.
    LaTeX is widely used by students and
    researchers because it allows easy formatting of sections, tables, and
mathematical expressions.

    \section{Methodology}
    In this document, LaTeX is used to demonstrate the basic structure of a
technical report.
    The document contains sections such as introduction, methodology, results, and
conclusion.
    The fancyhdr package is used to customize headers and footers.
```

```
\section{Results}
Using LaTeX improves the quality and organization of technical documents.
It automatically
manages section numbering, page formatting, and document structure, making it
easier to
produce professional reports.
```

```
\section{Conclusion}
LaTeX is a useful tool for preparing academic and technical documents.
It helps writers
focus on content while maintaining a clean and consistent layout throughout the
document.
```

```
\end{document}
```

Experiment 3: LaTeX Implementation Source Code

Below is the exact and unaltered LaTeX code structure with original indentation, packages, and commands intact:

```
\documentclass{article}
\usepackage{graphicx}
\usepackage{geometry}
\usepackage{setspace}

\geometry{
  a4paper,
  total={170mm,257mm},
  left=20mm,
  top=20mm,
}

\begin{document}

  \begin{titlepage}
    \centering
    \includegraphics[width=0.3\textwidth]{C:/Users/Vinyas K
S/OneDrive/Desktop/vtu logo.jpeg}
    \par
    {\LARGE \textbf{Visvesvaraya Technological University}\par}
    \vspace{1cm}
    {\Large Department of Computer Science and Engineering\par}
    \vspace{2cm}
    {\Huge\bfseries Technical Writing using LaTeX\par}
    \vspace{2cm}
    {\Large Submitted by:\par}
    {\large VINYAS K S\par}
    {\large 4VM24CS047\par}

    \vspace{1cm}
    {\Large Guide:\par}
    {\large Dhananjaya Kumar K\par}
    \vfill
    {\large IV SEMESTER\par}
  \end{titlepage}

\end{document}
```

Experiment 4: LaTeX Implementation Source Code

Below is the exact and unaltered LaTeX code structure with original indentation, packages, and commands intact:

```
\documentclass{article}
\usepackage[utf8]{inputenc}

\begin{document}

    \begin{center}
        \textbf{\LARGE Certificate}
    \end{center}

    \vspace{1cm}

    This is to certify that \underline{\textbf{Mr. Vinyas.K.S}} bearing USN:
    \underline{\textbf{4VM24CS047}} has been satisfactorily completed the course of
    experiment prescribed by the Visvesvaraya Technological University for the
    \underline{\textbf{4th}} semester Branch of Engineering degree during
    \underline{\textbf{2025-2026}}.

    \vspace{2cm}

    \begin{flushright}
        Date: \underline{\textbf{\today}}

        \par
        \vspace{1cm}

        \underline{\textbf{Dhananjaya Kumar K}}

        \par
        \textit{(Signature of Supervisor)}
    \end{flushright}

\end{document}
```

Experiment 5: LaTeX Implementation Source Code

Below is the exact and unaltered LaTeX code structure with original indentation, packages, and commands intact:

```
\documentclass{article}
\usepackage{multirow}

\begin{document}

    \begin{table}
        \caption{Student Result}
        \centering

        \begin{tabular}{|c|c|c|c|c|c|}
            \hline
            \multirow{2}{*}{SL NO} & \multirow{2}{*}{USN} & & & & \\
            \multirow{2}{*}{Student Name} & \multicolumn{3}{c|}{Marks} & & \\

            \cline{4-6}
            & & DBMS & LaTeX & Microcontrollers & \\
            \hline

            1 & 4VM24CS001 & Vinyas K S & 89 & 60 & 90 \\
            \hline

            2 & 4VM24CS002 & M J Manjunath & 78 & 47 & 98 \\
            \hline

            3 & 4VM24CS003 & Rakshitha CR & 100 & 69 & 59 \\
            \hline

        \end{tabular}
    \end{table}

\end{document}
```

Experiment 6: LaTeX Implementation Source Code

Below is the exact and unaltered LaTeX code structure with original indentation, packages, and commands intact:

```
\documentclass{article}
\usepackage{graphicx}
\usepackage{subcaption}

\begin{document}

    \begin{figure}[h!]
        \centering

        \begin{subfigure}[b]{0.4\linewidth}
            \includegraphics[width=\linewidth]{C:/Users/Vinyas K
S/OneDrive/Desktop/images.jpg}
            \caption{Caption for Image 1}
            \label{fig:img1}
        \end{subfigure}
        \hspace{0.1\linewidth}
        \begin{subfigure}[b]{0.4\linewidth}
            \includegraphics[width=\linewidth]{C:/Users/Vinyas K
S/OneDrive/Desktop/images (1).jpg}
            \caption{Caption for Image 2}
            \label{fig:img2}
        \end{subfigure}

        \caption{Main caption for both images}
        \label{fig:both_images}

    \end{figure}

\end{document}
```

Experiment 7: LaTeX Implementation Source Code

Below is the exact and unaltered LaTeX code structure with original indentation, packages, and commands intact:

```
\documentclass{article}
\usepackage{amsmath}

\begin{document}

    \title{Mathematical Equations}
    \author{VINYAS K S}
    \date{\today}
    \maketitle

    \section*{Equations}

    \subsection*{Equation 1}
    The quadratic equation is:

    \begin{equation}
        x=\frac{-b \pm \sqrt{b^2-4ac}}{2a}
        =\frac{-2 \pm \sqrt{2^2-4(1)(-8)}}{(2)(1)}
        =\frac{-2 \pm \sqrt{4+32}}{2}
    \end{equation}

    where  $(a,b,c)$  are the coefficient values.
    \subsection*{Equation 2}
    The second equation is:

    \begin{equation}
        \frac{d^2y}{dx^2}+\omega^2 y=0
    \end{equation}

    where  $(y(x))$  is a function of  $(x)$  and  $(\omega)$  is a constant.

\end{document}
```


Experiment 8: LaTeX Implementation Source Code

Below is the exact and unaltered LaTeX code structure with original indentation, packages, and commands intact:

```
\documentclass{article}
\usepackage{amsthm}

% Define theorem-like environments
\newtheorem{theorem}{Theorem}[section]
\newtheorem{definition}[theorem]{Definition}
\newtheorem{corollary}[theorem]{Corollary}
\newtheorem{lemma}[theorem]{Lemma}

\begin{document}

    \section{Introduction}

    In this section, we will present some theorems, definitions, corollaries, and
    lemmas.
    \section{Theorems and Definitions}

    \begin{theorem}
        This is a theorem.
    \end{theorem}

    \begin{definition}
        This is a definition.
    \end{definition}

    \begin{theorem}
        This is another theorem.
    \end{theorem}

    \begin{definition}
        This is another definition.
    \end{definition}

    \section{Corollaries and Lemmas}

    \begin{corollary}
        This is a corollary.
    \end{corollary}

    \begin{lemma}
        This is a lemma.
    \end{lemma}

    \begin{corollary}
```

This is another corollary.
 \end{corollary}

 \begin{lemma}
 This is another lemma.
 \end{lemma}

\end{document}

Experiment 9: LaTeX Implementation Source Code

Below is the exact and unaltered LaTeX code structure with original indentation, packages, and commands intact:

```
\documentclass{article}

\begin{document}

    \title{Introduction to Latex}
    \author{Dhananjaya Kumar K}
    \date{\today}
    \maketitle

    \section{Introduction}

    LaTeX is a document preparation system used for creating professional and
    scientific documents.
    It is widely used by students, researchers, and teachers for writing reports,
    articles, and project documentation \cite{citation1}, \cite{citation2},
    \cite{citation3}, \cite{citation4}, \cite{citation5}.
    LaTeX also provides support for managing citations and references in academic
    writing.
    It helps users maintain proper formatting and improves the quality of documents
    \cite{citation6}, \cite{citation7}, \cite{citation8}, \cite{citation9},
    \cite{citation10}.
    \section{References}

    \begin{thebibliography}{99}

        \bibitem{citation1}
        K Dhananjaya Kumar, J Akash. (2024). Introduction to Latex.
        \emph{IJRRT}, \textbf{Vol 3}(Issue 4), 45-56.

        \bibitem{citation2}
        Author2, B. (Year).
        Title of the paper.
        \emph{Journal Name}, \textbf{Volume}(Issue), page range.

        \bibitem{citation3}
        Author3, C. (Year). Title of the paper.
        \emph{Journal Name}, \textbf{Volume}(Issue), page range.

        \bibitem{citation4}
        Author4, D. (Year). Title of the paper.
        \emph{Journal Name}, \textbf{Volume}(Issue), page range.

        \bibitem{citation5}
        Author5, E. (Year).
```

Title of the paper.
`\emph{Journal Name}, \textbf{Volume}(Issue), page range.`

`\bibitem{citation6}`
Author6, F. (Year). Title of the paper.
`\emph{Journal Name}, \textbf{Volume}(Issue), page range.`

`\bibitem{citation7}`
Author7, G. (Year). Title of the paper.
`\emph{Journal Name}, \textbf{Volume}(Issue), page range.`

`\bibitem{citation8}`
Author8, H. (Year).
Title of the paper.
`\emph{Journal Name}, \textbf{Volume}(Issue), page range.`

`\bibitem{citation9}`
Author9, I. (Year). Title of the paper.
`\emph{Journal Name}, \textbf{Volume}(Issue), page range.`

`\bibitem{citation10}`
Author10, J. (Year). Title of the paper.
`\emph{Journal Name}, \textbf{Volume}(Issue), page range.`

`\end{thebibliography}`

`\end{document}`

Experiment 10: LaTeX Implementation Source Code

Below is the exact and unaltered LaTeX code structure with original indentation, packages, and commands intact:

```
\documentclass{article}
\usepackage{tikz}

\begin{document}

    \begin{tikzpicture}[
        level 1/.style={sibling distance=4cm},
        level 2/.style={sibling distance=2.5cm},
        every node/.style={draw, circle, minimum size=1cm}
    ]

        \node {Root}
        child {node {Child 1}
            child {node {GC 1}}
            child {node {GC 2}}
        }
        child {node {Child 2}
            child {node {GC 3}}
            child {node {GC 4}}
        };

    \end{tikzpicture}

\end{document}
```

Experiment 11: LaTeX Implementation Source Code

Below is the exact and unaltered LaTeX code structure with original indentation, packages, and commands intact:

```
\documentclass{article}
\usepackage{algorithm}
\usepackage{algorithmic}

\begin{document}

  \section{Algorithm using \texttt{algorithm} and \texttt{algorithmic} Packages}

  \begin{algorithm}
    \caption{Example Algorithm}\label{alg:example1}

    \begin{algorithmic}[1]

      \REQUIRE $n \geq 0$
      \ENSURE $y = x^n$

      \STATE $y \gets 1$
      \STATE $x \gets x_0$
      \STATE $n \gets n_0$

      \WHILE{$n \neq 0$}

        \IF{$n$ is even}
          \STATE $x \gets x \times x$
          \STATE $n \gets n / 2$
        \ELSE
          \STATE $y \gets y \times x$
          \STATE $n \gets n - 1$
        \ENDIF

      \ENDWHILE

    \end{algorithmic}
  \end{algorithm}

\end{document}
```

Experiment 12: LaTeX Implementation Source Code

Below is the exact and unaltered LaTeX code structure with original indentation, packages, and commands intact:

```
\documentclass{report}
\title{MINI PROJECT REPORT}
\author{VINYAS K S}
\date{\today}
\begin{document}
  \maketitle
  \tableofcontents
  \chapter{Introduction}
  This is the introduction of my simple report.
  \section{Background}
  Here, I'll provide some background information.
  \chapter{Methods}
  This chapter describes the methods used in my study.
  \chapter{Results}
  Here are the results of my study.
  \chapter{Conclusion}
  In conclusion, I'll summarize my findings.
\end{document}
```

```
\documentclass{article}
\title{My Simple Article}
\author{VINYAS K S}
\date{\today}
\begin{document}
  \maketitle
  \section{Introduction}
  This is the introduction of my simple article.
  \subsection{Background}
  Here, I'll provide some background information. \section{Methods}
  This section describes the methods used in my study.
  \section{Results}
  Here are the results of my study.
  \section{Discussion}
  I'll discuss the implications of the results in this section.
  \section{Conclusion}
  In conclusion, I'll summarize my findings.
\end{document}
```